Simple Harmonic Motion Calculations: A Beginner's Guide

Lesson 4: Calculating Displacement, Velocity, and Acceleration

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Calculating Displacement, Velocity, and Acceleration - Study Notes

Key Concepts

- **Displacement (x(t)):** The position of an object at a given time. Represented by the equation x(t) = Acos(<—B ² Æ).
- **Velocity (v(t)):** The rate of change of displacement with respect to time (the derivative of x(t)).
- **Acceleration (a(t)):** The rate of change of velocity with respect to time (the derivative of v(t)). Also related to displacement.
- **Derivatives:** Understanding how to find the derivative of a function is crucial for calculating velocity and acceleration.
- **Equilibrium Position:** The position where the object naturally rests (in this case, x=0 when <cÓ ’à

Summary

This lesson focuses on how to calculate velocity and acceleration from a given displacement equation, specifically x(t) = Acos(<—B ² Æ). Velocity is found by taking the first derivative of the displacement equation with respect to time, resulting in v(t) = -A<—6–âƒÉt + <b’à Similarly, acceleration is found by taking the first derivative of the velocity equation, yielding a(t) = -A<›&6÷2ƒÉt + <b’à A particularly important result is that acceleration is directly proportional to displacement and always points towards the equilibrium position (a(t) = -<›',‡B’’à

The relationship a(t) = -<›',‡B’ †–v†Æ–v‡G2   °ey characteristic of oscillatory motion: when the displacement is positive, the acceleration is negative (pulling the object back towards equilibrium), and when the displacement is negative, the acceleration is positive (also pulling the object back towards equilibrium). This constant restoring force is what drives the oscillation. Understanding these relationships is fundamental to analyzing and predicting the behavior of oscillating systems.

Important Points

- **x(t) = Acos(<—B ² Æ):** This equation describes simple harmonic motion.
   *   `A`: Amplitude (maximum displacement from equilibrium)
   *   `<–   Angular frequency (related to the frequency and period of the oscillation)
   *   `<f   Phase constant (determines the initial position of the object)
- **v(t) = -A<—6–âƒÉt + <b“¢¢¢ Velocity is the derivative of displacement. The negative sign indicates the velocity is opposite in direction to the displacement.
- **a(t) = -A<›&6÷2ƒÉt + <b“¢¢¢ 66VÆW ation is the derivative of velocity.
- **a(t) = -<›',‡B“¢¢¢   7 ucial relationship showing acceleration is proportional to displacement and always points towards the equilibrium position.
- **Derivatives are Key:** Velocity and acceleration are defined as the rate of change of displacement and velocity, respectively. This requires understanding basic calculus (derivatives).

- **Units:** Ensure consistent units are used throughout calculations (e.g., meters for displacement, meters/second for velocity, meters/second^2 for acceleration, radians/second for angular frequency).

Examples

Example 1: Finding Displacement, Velocity, and Acceleration

A mass oscillates with A = 0.2 m, <' Ò R ad/s, and <b Ò À/2. Find the displacement, velocity, and acceleration at t = 0.2 s.

- **Displacement (x(t)):**

  x(0.2) = 0.2 * cos(5 * 0.2 + < ó"' Ò ã" ¢ 6÷2ƒ ^{2} À/2) "H 0.2 * cos(2.57) "H 0.2 * (-0.88) "H -0.176 m

- **Velocity (v(t)):**

  v(0.2) = -0.2 * 5 * sin(5 * 0.2 + < ó"' Ò Ó ¢ 6–âƒ ^{2} À/2) "H -1 * sin(2.57) "H -1 * (-0.99) "H 0.99 m/s

- **Acceleration (a(t)):**

  a(0.2) = -5^2 * cos(5 * 0.2 + < ó"' Ò Ó#R ¢ 6÷2ƒ ^{2} À/2) "H -25 * (-0.88) "H 22 m/s^2

**Example 2: Using a(t) = -<›',‡B'¢

Using the same values as Example 1 (A = 0.2 m, <' Ò R ad/s, <b Ò À/2), let's calculate acceleration at t = 0.2 s using the simplified equation:

- **Displacement at t=0.2s (from Example 1):** x(0.2) "H -0.176 m
- **Acceleration (a(t)):**

  a(0.2) = -<›" ¢ ,ƒ ã"' Ò Òƒ["' ¢ ,Ó ã sb' Ò Ó#R ¢ ,Ó ã sb' $, BãB Ò÷;

Note: There's a slight discrepancy between the two acceleration calculations due to rounding errors. The a(t) = -<›',‡B' ÖWF†öB —2 vVæW ally more accurate.

Quick Review

1. What is the derivative of x(t) = Acos(<—B ^{2} Æ) with respect to time?
2. What does the phase constant (<b' &W &W6VçB –â F†R F—7 Æ 6VÖVçB W V F–öãð
3. Explain why the acceleration in simple harmonic motion always points towards the equilibrium position.
4. If the amplitude (A) of an oscillation is doubled, how does this affect the maximum velocity?
5. What is the relationship between acceleration and displacement in simple harmonic motion, as expressed by the equation a(t) = -<›',‡B"ð

Further Reading

- **Simple Harmonic Motion:** Explore the properties of simple harmonic motion in more detail, including period, frequency, and energy.
- **Derivatives in Physics:** Review the concept of derivatives and their applications in physics, particularly in kinematics.
- **Pendulum Motion:** Investigate how the principles of simple harmonic motion apply to the motion of a pendulum (for small angles).
- **Damped Oscillations:** Learn about what happens when energy is lost from an oscillating system, leading to damped oscillations.
- **Driven Oscillations:** Explore what happens when an oscillating system is subjected to an external force.